Extending the Notebooks

Rotation, electromagnetism, Ricci-flat spaces, two scale factors, and matter and radiation

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Purpose and conventions

These notes collect metrics and equations for the next steps listed in Section 8 of the Introduction and Summary, written in the same conventions as the notebooks so that they can be entered directly.

- **Signature** (+, −, −, −, …): $g_{00} > 0$, spatial terms negative.
- **Ricci tensor** as defined in the notebooks, $R_{\mu\nu} = \partial_\nu \Gamma^\sigma_{\mu\sigma} - \partial_\sigma \Gamma^\sigma_{\mu\nu} + \Gamma^\rho_{\mu\sigma}\Gamma^\sigma_{\rho\nu} - \Gamma^\rho_{\mu\nu}\Gamma^\sigma_{\rho\sigma}$. This is the negative of the definition in many textbooks; combined with the (+, −, −, −) signature, the vacuum equations take the form $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$ used throughout.
- **Units:** G and c are kept. Lengths built from mass, angular momentum and charge are $m = GM/c^2$, $a = J/(Mc)$, and $q^2 = GQ^2/(4\pi\varepsilon_0 c^4)$.
- **Λ in D dimensions:** the notebooks write $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$ in every dimension. The more common convention, $R_{\mu\nu} - \tfrac{1}{2}R\, g_{\mu\nu} + \Lambda_{std}\, g_{\mu\nu} = \kappa T_{\mu\nu}$, gives in vacuum $R_{\mu\nu} = 2\Lambda_{std}/(D-2)\, g_{\mu\nu}$. So the notebooks' Λ equals $2\Lambda_{std}/(D-2)$; the two agree in 4D.

All formulas marked 'checked' were verified by Claude with SymPy against the notebooks' Ricci definition. The others are standard results quoted for orientation.

1. Matter and radiation

With matter, the vacuum equations $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$ become, in D spacetime dimensions (checked for D = 4, 5 and 7):

$$R_{\mu\nu} = \Lambda\, g_{\mu\nu} - \kappa\,[\, T_{\mu\nu} - T\, g_{\mu\nu}/(D-2)\,], \quad \kappa = 8\pi G/c^4,$$

where $T_{\mu\nu}$ is the stress-energy tensor and $T = g^{\mu\nu}T_{\mu\nu}$ its trace. For a perfect fluid with density ρ (mass per volume), pressure p, and four-velocity $u^\mu = (1, 0, 0, \ldots)$ in the notebooks' comoving coordinates (so $u_\mu = (c^2, 0, 0, \ldots)$):

$$T_{\mu\nu} = (\rho + p/c^2)\, u_\mu u_\nu - p\, g_{\mu\nu}.$$

For the 4D GRW metric ($A = c^2$) these give the standard Friedmann equations (checked):

$$(\dot a/a)^2 = (8\pi G/3)\,\rho + \Lambda c^2/3 - kc^2/a^2, \qquad \ddot a/a = -(4\pi G/3)(\rho + 3p/c^2) + \Lambda c^2/3,$$

together with the continuity equation $\dot\rho = -3(\dot a/a)(\rho + p/c^2)$. Setting ρ = p = 0 recovers the notebooks' result $\dot a^2 = \Lambda c^2 a^2/3 - kc^2$. With n spatial dimensions sharing one scale factor (checked for n = 3, 4, 6, k = 0):

$$\tfrac{1}{2}\, n(n-1)\,[\,(\dot a/a)^2 + kc^2/a^2\,] = 8\pi G_D\,\rho + \tfrac{1}{2}\,(n-1)\,\Lambda c^2, \qquad \dot\rho = -n\,(\dot a/a)(\rho + p/c^2).$$

Content	Equation of state	Density, n spatial dims	Density, 3 spatial dims
Dust (matter)	$p = 0$	$\rho \propto a^{-n}$	$\rho \propto a^{-3}$
Radiation	$p = \rho c^2/n$	$\rho \propto a^{-(n+1)}$	$\rho \propto a^{-4}$
Vacuum energy	$p = -\rho c^2$	ρ constant (acts like Λ)	ρ constant

A caution for the higher-dimensional cases: ordinary matter and radiation live in the three familiar dimensions and do not push on the curled ones in the same way. The pressure is then different in the two kinds of direction, and one shared scale factor is no longer consistent. Adding matter therefore goes together with the two scale factors of Section 3.

2. Rotation: the Kerr–de Sitter metric

Rotation requires an off-diagonal term $dt\,d\varphi$, and the functions then depend on r and θ. The exact rotating vacuum solution with Λ in 4D is the Kerr–de Sitter metric (Carter 1968). In Boyer–Lindquist coordinates and the notebooks' signature (checked):

$$ds^2 = (\Delta_r/\rho^2)(c\,dt - (a\sin^2\theta/\Xi)\,d\varphi)^2 - (\Delta_\theta \sin^2\theta/\rho^2)(a\,c\,dt - ((r^2+a^2)/\Xi)\,d\varphi)^2 - (\rho^2/\Delta_r)\,dr^2 - (\rho^2/\Delta_\theta)\,d\theta^2,$$

$$\rho^2 = r^2 + a^2\cos^2\theta, \quad \Delta_r = (r^2+a^2)(1-\Lambda r^2/3) - 2mr, \quad \Delta_\theta = 1 + (\Lambda a^2/3)\cos^2\theta, \quad \Xi = 1 + \Lambda a^2/3.$$

- $a = 0$ gives the Schwarzschild–de Sitter metric, with $g_{00} = c^2(1 - 2m/r - \Lambda r^2/3)$.
- $\Lambda = 0$ gives the Kerr metric of a rotating black hole.
- Adding q^2 to Δ_r gives the charged, rotating Kerr–Newman–de Sitter metric. It is no longer a vacuum solution: it needs the electromagnetic stress tensor of Section 4 on the right-hand side.

This is a rotating black hole in an expanding (de Sitter) universe, not a rotating universe. For a first look at rotation within the notebooks' cosmological setting, the simplest ansatz adds one small off-diagonal term to a GRW metric:

$$ds^2 = (\text{GRW line element}) + 2\,a(t)^2\,r^2\sin^2\theta\,\Omega(t, r)\,c\,dt\,d\varphi,$$

keeping only terms of first order in Ω. The new field equation is the off-diagonal $R_{03} = 0$ ($R_{03} = \Lambda g_{03}$ to this order), which the notebooks' off-diagonal machinery already handles. In higher dimensions the rotating black holes are the Myers–Perry solutions.

3. Two scale factors and Ricci-flat compact spaces

Give ordinary space and the curled space their own scale factors:

$$ds^2 = c^2\,dt^2 - a(t)^2\,h_3 - b(t)^2\,h_{int},$$

where h_3 is a 3-space of constant curvature k and h_{int} is a compact d-dimensional space with zero Ricci curvature. The vacuum equations $R_{\mu\nu} = \Lambda\,g_{\mu\nu}$ reduce to three ordinary differential equations (checked for $d = 2$):

$$3\,\ddot{a}/a + d\,\ddot{b}/b = \Lambda c^2$$

$$\ddot{a}/a + 2(\dot{a}/a)^2 + d\,(\dot{a}/a)(\dot{b}/b) + 2kc^2/a^2 = \Lambda c^2$$

$$\ddot{b}/b + (d-1)(\dot{b}/b)^2 + 3\,(\dot{a}/a)(\dot{b}/b) = \Lambda c^2$$

Example.

Start from the 6D_1 notebook, whose curled space is the torus $T^2 = S^1 \times S^1$, a compact 2-dimensional Ricci-flat space (d = 2). Give the curled space its own scale factor b(t), set $A = c^2$ and $D_4 = D_5 = 1$, and leave B and C free. The field equations then give $B = (2C + rC')^2/[4C(1 - kr^2C)]$ with C(r) free, together with the three equations above. This is carried out in the notebook 6D_1 Two Scale Factors, and for six circles (d = 6) in 10D_1; see the folder '6 Toward Calabi-Yau'.

With b = a, d circles and k = 0 these reduce to the notebooks' single-scale-factor case. A solution in which b stays constant while a grows is what a realistic model needs. In vacuum the third equation then requires Λ = 0, which is why stabilizing the curled space needs extra ingredients such as matter, fluxes, or a potential.

Calabi–Yau spaces.

A Calabi–Yau space is not one space but a large class: compact complex spaces of a particular type that admit a Ricci-flat metric, as Yau proved in 1977 (the Calabi conjecture). The six-dimensional ones used in string theory number at least in the tens of thousands of distinct types. No explicit formula for the Ricci-flat metric is known for any nontrivial example; Yau's theorem proves existence only. They therefore cannot be entered into the notebooks the way a circle or a sphere can. They do not need to be, however: in the equations above only the Ricci tensor of h_{int} enters, and it is zero. Any Ricci-flat compact space, a torus or a Calabi–Yau, gives the same three equations with d its dimension (d = 6 for string theory's Calabi–Yau spaces), and none of them triggers the curvature test's closed-universe conclusion.

4. Electromagnetism

There are two routes.

(a) Einstein–Maxwell in 4D.

Use the equations of Section 1 with the electromagnetic stress tensor (SI units, notebooks' signature)

$$T^{\mu\nu} = (1/\mu_0)\,[\,F^{\mu\alpha}F_{\alpha}{}^{\nu} + \tfrac{1}{4}\,g^{\mu\nu}\,F_{\alpha\beta}F^{\alpha\beta}\,],$$

which is traceless in 4D, so the field equations become $R_{\mu\nu} = \Lambda\, g_{\mu\nu} - \kappa\, T_{\mu\nu}$. The simplest test case keeps a diagonal metric: the charged, non-rotating Reissner–Nordström–de Sitter solution,

$$ds^2 = f\,c^2\,dt^2 - dr^2/f - r^2(d\theta^2 + \sin^2\theta\,d\varphi^2), \quad f = 1 - 2m/r + q^2/r^2 - \Lambda r^2/3,$$

with an electric potential proportional to Q/r. Since the metric is diagonal and depends only on r, the notebooks can compute its Ricci tensor unchanged; only the right-hand side is new.

(b) Kaluza–Klein: electromagnetism from a curled circle.

No new equations are needed. Add off-diagonal terms between ordinary spacetime and the circle ψ, and keep the vacuum equations $R_{AB} = \Lambda g_{AB}$ in five dimensions:

$$ds^2 = g_{\mu\nu}\, dx^\mu\, dx^\nu - \varphi^2\, (p\, d\psi + \kappa_5\, A_\mu\, dx^\mu)^2,$$

where A_μ plays the role of the electromagnetic potential, φ is a scalar field (the size of the circle), and κ_5 is a coupling constant. The simplest example is a static electric field, $A = A_0(r)\, dt$, which adds a single off-diagonal term $g_{0\psi}$. The five-dimensional equations then contain the four-dimensional Einstein–Maxwell equations together with an equation for φ. For $\Lambda = 0$ exact charged solutions are known (for example by 'boosting' the Schwarzschild metric times a circle along ψ; Gibbons and Wiltshire 1986), which would make a good check of a modified notebook.

References

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2. R. C. Myers and M. J. Perry, "Black holes in higher dimensional space-times," Ann. Phys. 172, 304 (1986).
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